



University of Venda

FORMAL REQUIREMENTS DOCUMENT

INTERN REGISTER SYSTEM

DOCUMENT VERSION: 1.0

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PREPARED FOR: UNIVERSITY OF VENDA (UNIVEN)

PREPARED BY: INTERN DEVELOPMENT TEAM

Document Control

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Acronym:

ICT: Information Communication Technology

VC: Vice-Chancellor and Principal

PIN: Personal Identification Number

EDRMS: Electronic Document & Records Management System

DOB: Date of Birth

HEIs: Higher Education Institutions

SA: South Africa

HR: Human Resources

API: Application Programming Interface

ITS: Enterprise Resource System name

RBAC: Role-Based Access Control

Definitions:

System / Tool: The proposed IRMS solution.

Intern: Student/graduate required to record attendance and request leave.

Supervisor: Staff member assigned to a Field who oversees interns.

Admin: Department-level administrator who manages Supervisors and Fields.

Super Admin: University-level administrator who manages Admins and Departments.

Users: Authorized officials permitted to use the system.

System Owner: Directorate/division responsible for the process and system.

Repository: Central storage for documents/records (e.g., EDRMS).

Integration: Data exchange between IRMS and institutional systems (e.g., ITS).

Interface: Any functionality/screen exposed to users.

1. Purpose

The purpose of the Intern Register Management System (IRMS) is to provide a centralized, digital platform for managing intern daily attendance, leave requests, and reporting functions across all University departments. The system aims to eliminate manual, paper-based processes; improve data accuracy and auditability; and support the University's strategic objective of institutional digital transformation through automated, secure, and role-based workflows.

2. Objective

The objective of this Software Requirements Document is to define the proposed solution for enhancing and modernizing intern attendance and leave management processes at the University of Venda. The current manual workflow—dependent on paper logbooks, physical attendance registers, and fragmented consolidation efforts—has resulted in operational inefficiencies for Human Resources personnel and Supervisors, particularly during payroll preparation cycles. Recognizing the broader institutional shift toward digital transformation, this project aims to introduce an automated, geolocation-enabled digital attendance platform that improves accuracy, reduces administrative workload, and establishes consistent, auditable processes across all departments. The Intern Register Management System (IRMS) is designed to achieve the following objectives:

- a) **Digitize and automate attendance management** by replacing traditional paper-based logbooks with a secure digital platform.
- b) **Improve service delivery**, ensuring timely stipend processing through accurate and real-time attendance data.
- c) **Simplify administrative tasks** by automating calculations related to attendance, hours worked, and summaries required for payroll.
- d) **Eliminate errors and reduce fraudulent attendance claims** through geolocation verification and validated sign-in/out workflows.

- e) **Enable the generation of mandatory reports**, including monthly timesheets and departmental summaries, in standardized PDF or Excel formats.
- f) **Maintain complete audit trails** for compliance, historical reference, and internal or external review.
- g) **Streamline reporting processes** by providing supervisors, administrators, and HR personnel with accessible, role-based dashboards and data exports.

Delivering high-quality, efficient service to all stakeholders remains a core priority for the University of Venda. The implementation of a digital intern management system directly supports this priority by modernizing outdated manual processes. Both internal research findings and the Business Systems division have emphasized the urgency of adopting the IRMS to resolve current administrative challenges, enhance transparency, and ensure operational efficiency across the institution.

3. Background

Based on comprehensive engagements with representatives from the Human Resources (HR) and Student Affairs departments, a critical business case was presented during the requirements gathering phase. These departments, which are responsible for the management and stipend processing of hundreds of university interns, expressed an urgent need for an electronic system to digitize the currently manual attendance tracking process.

The current system relies heavily on physical paper logbooks, where interns manually sign in and out daily. This traditional method has proven to be:

- a) **Time-consuming:** HR staff spend manually days calculating hours from hundreds of logbooks to process monthly stipends.
- b) **Ineffective:** Physical logbooks are often lost, damaged, or submitted late, delaying payments.

- c) **Prone to Errors and Fraud:** The manual system is susceptible to "buddy punching" (signing in for a friend) and calculation errors, leading to financial discrepancies.
- d) **Lacking Visibility:** Supervisors cannot monitor attendance in real-time, leading to "ghost workers" who claim stipends without being physically present.

In the current digital era, such inefficiencies are unsustainable. The transformation of these conventional methods into a fully automated digital solution is not just desired but necessary. This automation initiative aligns directly with the instructions of the Vice-Chancellor and Principal, as outlined in the University's Strategic Plan for 2021-2025. Specifically, Strategic Objective No. 12 emphasizes the urgency of automation and digitalization of business processes to improve efficiency and governance. To ensure the provision and maintenance of quality services, all directorates, including HR and Student Affairs, are mandated to fulfill these strategic initiatives.

4. User stories

The Univen Intern Register System provides tailored functionality for Interns, Supervisors, Admins, and Super Admins. Interns require secure access and account control, so they must be able to register using their university email, log in with their credentials, reset forgotten passwords through email verification, and view their profile details such as department, supervisor, and contract end date to ensure accuracy. Supervisors need to be automatically linked to their correct department so they can oversee the appropriate group of interns. Administrators must manage user accounts by activating or deactivating them to maintain system security when staff changes occur. At a higher level, Super Admins need the ability to register and manage Admin accounts so that responsibilities can be delegated, as well as configure system-wide settings such as global leave policies to align the system with university regulations.

Attendance management focuses heavily on intern activities. Interns must be able to sign in daily from their devices with automatic GPS verification to confirm physical

presence at the workplace. They also need to sign out at the end of each day to ensure correct calculation of working hours, digitally sign their attendance records for payroll validity, and access their attendance history for reference. Supervisors require real-time notifications whenever their interns sign in, along with the ability to view a list of all present interns on a given day and see the GPS location of each sign-in to verify that interns are at the correct site.

Leave management supports both interns and supervisors. Interns need to submit leave requests whether sick, study, annual, or family leave and attach supporting documents such as medical certificates. They must also be able to view the status of their requests to see whether they have been approved or declined. Supervisors and admins should be able to view pending leave requests, approve or reject them, and provide comments when declining so interns understand the reason for the decision.

For reporting and analytics, Supervisors and admins require the ability to generate monthly timesheets for interns and download PDF/Excel reports for printing and filing. Admins need a dashboard summarizing the total number of interns per department, along with the option to export all attendance data to Excel /PDF for deeper analysis or payroll integration. Super Admins should be able to create Admin accounts and assign Departments.

5.Functional and Non-functional Requirements

5.1. Functional Requirements

5.1.1. BUSINESS RULE

- The system should be web-based and mobile-responsive.
- Only authorized users (Intern, Supervisor, Admin, Super Admin) will utilize the system; access is RBAC.
- After login, the user's role and permitted functions auto-load.
- Interns' self-register; Admins are created by Super Admin and assigned to Departments; Supervisors are created by Admin and assigned to Fields.
- Admin and Supervisors approve/decline leave; declines must include a reason; all decisions are audited.
- Admins manage Supervisors and Fields within their assigned Department.
- Super Admins manage Admins and Departments (university-wide).
- The system supports exports (PDF/Excel) and integration with ITS/EDRMS.

5.1.2. User Interface

(a) Create Account: Intern Self-Registration

Create Your Account
Join the University of Venda Intern Program

Secure registration process for interns only

First Name * **Surname *** **ID Number *** **Staff Email ***

Enter your first name Enter your surname Enter your 13-digit ID 13 digits their.email@univen.ac.za Send Code
Must be a valid @univen.ac.za email address

User Role * **Verification Code ***

Intern Enter 6-digit code

Employer Name * **Password ***

Enter your employer's name Create a strong password
Required for intern registration

Start Date * **End Date ***

mm/dd/yyyy mm/dd/yyyy
Internship start date Internship end date

Department * **Confirm Password ***

Select your department Re-enter your password

Field *

Select department first
Please select a department first to see available fields

☐ I have read and agree to the [Terms and Conditions](#) *

[Go to Create Account](#)

Database Tables Used:

- users (create login) • role='INTERN', active initially FALSE/TRUE per policy
- verification_codes (send & confirm email code)
- terms_acceptance (record acceptance at sign-up)
- interns (create minimal profile; supervisor/department can be linked now or later)

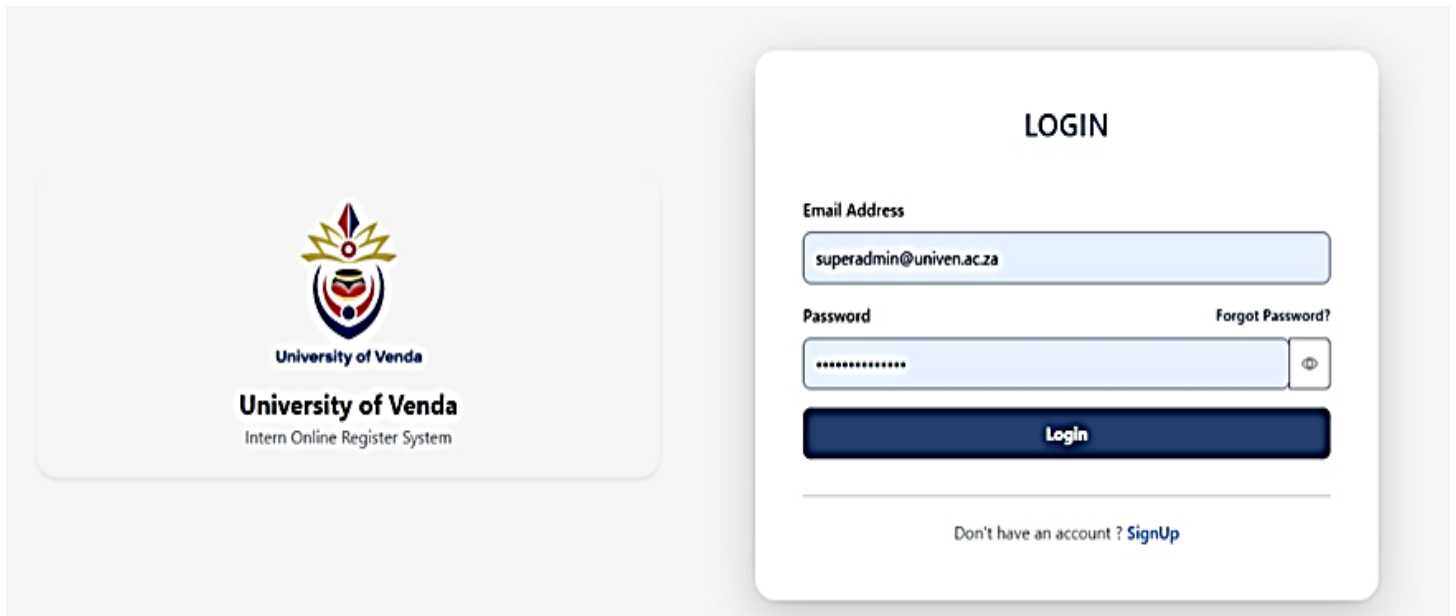
Fields to be used:

Field Name	Data Type	Description / Source
Full Names	Text	Creates interns.name and users.name (keep in sync)
Email	Text	Creates users.email and interns.email; used for verification code
Password / Confirm	Password	Saves hashed into users.password (role=INTERN)
Department	Select	Writes interns.department_id (or leave null until Admin assigns)
Field	Text/Select	Writes interns.field (text column in schema)
Supervisor (optional)	Select	Writes interns.supervisor_id (can be assigned later by Admin)
ID Number (optional)	Text	interns.id_number (if captured at sign-up)
Start / End Date (optional)	Date	interns.start_date, interns.end_date (or set later)
Accept Terms	Checkbox	Inserts terms_acceptance.accepted=TRUE, accepted_at, ip_address, version, user_id
Email Verification Code	Text	Validated against verification_codes.email, code, expires_at before activating login
Employer	Text	Name of the company the intern is employed with

NB:

- On submit, create **users'** row (role='INTERN', active per policy) then **interns** row with the same email/name; record **terms_acceptance**; verify code via **verification_codes** before first login.

(b) Login Interface



University of Venda
Intern Online Register System

LOGIN

Email Address
superadmin@univen.ac.za

Password
.....

[Forgot Password?](#)

Login

Don't have an account ? [SignUp](#)

Database Tables Used:

- users, verification_codes, password_reset_tokens, terms_acceptance

Fields to be used:

Field Name	Data Type	Description / Source
Username / Email	Text	users.username or users.email (credential)
Password	Password	users.password (hashed)
2FA / Code (optional)	Text	verification_codes.code, verification_codes.email, expires_at (if enabled)
Forgot Password	Link/Action	Issues password_reset_tokens.token, expires_at, marks used on completion
Accept Terms	Checkbox	Writes terms_acceptance.accepted, accepted_at, ip_address, version, user_id

c. Manage Interns Interface (Shared Interface)

This interface is used to view and manage intern profiles. Both the Admin and Supervisor utilize this interface to access intern details, but their rights differ based on their role.

NB:

- This interface does not allow the creation of interns.
- Intern creation occurs through Intern Self-Registration or back-office provisioning.
- Admins and Supervisors may view and update intern information in line with their permissions.
- The system must ensure that users only see interns relevant to their assigned departmental scope.

Interns Management

Name/Email

Search by name or email...

Department

All Departments

Field

All Fields

Status

All Status

Reset Filters

Showing 1 - 1 of 1 interns

Total: 1

Items per page: 25

Name	Email	ID Number	Start Date	End Date	Supervisor	Department	Field	Status	Actions
intern baloyi	intern.baloyi@univen.ac.za	1243678912431	2025-11-03	2026-10-30	muthuhadini Ramavhunga	ICT	software development	Present	 

« 1 »

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Fields to be used:

Field Name	Data Type	Description
Full Names	Text Input	Intern's names (interns.name)
Email	Text Input	Intern's email (interns.email)
Department	Auto/Lookup	interns.department_id → Departments
Field (Specialisation)	Text Input	interns.field
Supervisor	Auto/Lookup	interns.supervisor_id → Supervisors
Company Name of the Funder	Text Input	Sponsor company (interns.employer)
Start Date	Date Input	interns.start_date
End Date	Date Input	interns.end_date
Status	Toggle	Intern Active/Inactive (interns.active)

Supervisor Access

NB: Supervisor access is limited to interns assigned to their Field(s).

- View list of interns under their assigned field(s)
- Edit intern details limited to: Field, Funder/Company Name, Contract Dates, Status (activate/deactivate)
- Cannot view interns belonging to other fields
- Cannot create or delete interns

Admin Access

NB: Admin access applies to all interns within the Department.

- View list of all department interns
- Edit intern details including Supervisor assignment, Field, Funder/Company Name, Contract dates, Status (activate/deactivate)
- Cannot create or delete interns
- Cannot view interns outside their department

D, Attendance History Interface (Shared Interface: Interns, Supervisors, Admins)

- **Attendance Records Table**

Field Name	Data Type	Description
attendance_id	INT (PK, Auto)	Unique identifier for each attendance record
user_id	INT (FK)	Employee/Intern who the record belongs to
date	DATE	Attendance date
status	ENUM	(Present, Absent, Late, Excused)
time_in	TIME	Check-in time
time_out	TIME	Check-out time

NB:

- Interns: Can view **only their own** attendance history.
- Supervisors: Can view **attendance of interns they supervise**.
- Admins: Can view **all attendance records**.
- The system must include filtering tools (date range, status, intern name, role).

Interface Layout:

 **Attendance History**

View weekly attendance records and track intern attendance patterns over time.

Quick Select: 📅 This Week 📅 Last Week 📅 2 Weeks Ago

 **Week & Filters**

☐ Monday Date

☐ Friday Date

Intern Name

Department

Field



No Week Selected

Select a Monday and Friday date to view weekly attendance history, or use the quick select buttons above.

E. Attendance Report Generation Interface (Shared Interface: Supervisors, interns & Admins)

Report Generation Fields:

Field Name	Data Type	Description
report_id	INT (PK, Auto)	Unique report identifier
generated_by	INT (FK)	User generating the report
role_scope	ENUM	(Supervisor, Admin)
date_range_start	DATE	Start date for the report
date_range_end	DATE	End date for the report
group_selected	VARCHAR (255)	Department / Intern Group / Individual
report_type	ENUM	Summary or Detailed
created_at	TIMESTAMP	Timestamp for report generation

NB:

- Supervisors can generate reports **only for their supervised interns**.
- Admins can generate **system-wide reports**.
- interns **can generate** their own personal attendance report
- Reports must be exportable in **PDF, Excel, or CSV**.
- The system must allow filters such as:
 - Date range
 - Intern name
 - Department / Project
 - Attendance status

Interface Layout:

1

Total Days

0

Present

0

Absent

0%

Attendance Rate

Quick Date Range Presets

Today

This Week

This Month

Quarter

Year

From Date *

mm/dd/yyyy

Select the start date for the report

To Date *

12/dd/2025

Select the end date for the report

Report Options

☐ Include Status

☒ Include Signatures

Generate Report

Reset

Export Report

Generate a report first to enable download options

Download PDF Report

Download Excel Report

Report Preview

Date	Day	Time In	Time Out	Duration	Location	Signature
Dec 03, 2025	Wednesday	-	08:39	-	testing	

F. Leave Management Interface (Shared Interface)

- This interface allows the Admin and Supervisor to approve or decline leave requests submitted by interns.

Interface Layout:

Debug Info: Active Section: Intern Leave status | Total Requests: 2 | Filtered: 2 | Paginated: 2 | Current Page: 1 | Total Pages: 1

Intern Leave Requests Test Alert Reset Seen Reload Data

Intern Name: Department: Field: Reset Filters

Showing 1 - 2 of 2 requests Total: 2 Items per page: 25

INTERN	EMAIL	DEPARTMENT	FIELD	START DATE	END DATE	REASON	STATUS	DOCUMENTS	ACTIONS
intern baloyi	intern.baloyi@univen.ac.za	ICT	software development	2025-12-04	2025-12-11		Approved		
intern baloyi	intern.baloyi@univen.ac.za	ICT	software development	2025-12-04	2025-12-11	no attachment	Declined	-	

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Report Generation Fields:

Field Name	Data Type	Description
Leave Type	Selection	leave_requests.leave_type
Start Date	Date	leave_requests.from_date
End Date	Date	leave_requests.to_date
Reason	Text Input	leave_requests.reason
Attachment	File Upload	leave_requests.attachment_path
Status	Auto	Pending / Approved / Declined
Rejection Comment	Text Input	leave_requests.rejection_message / decline_reason

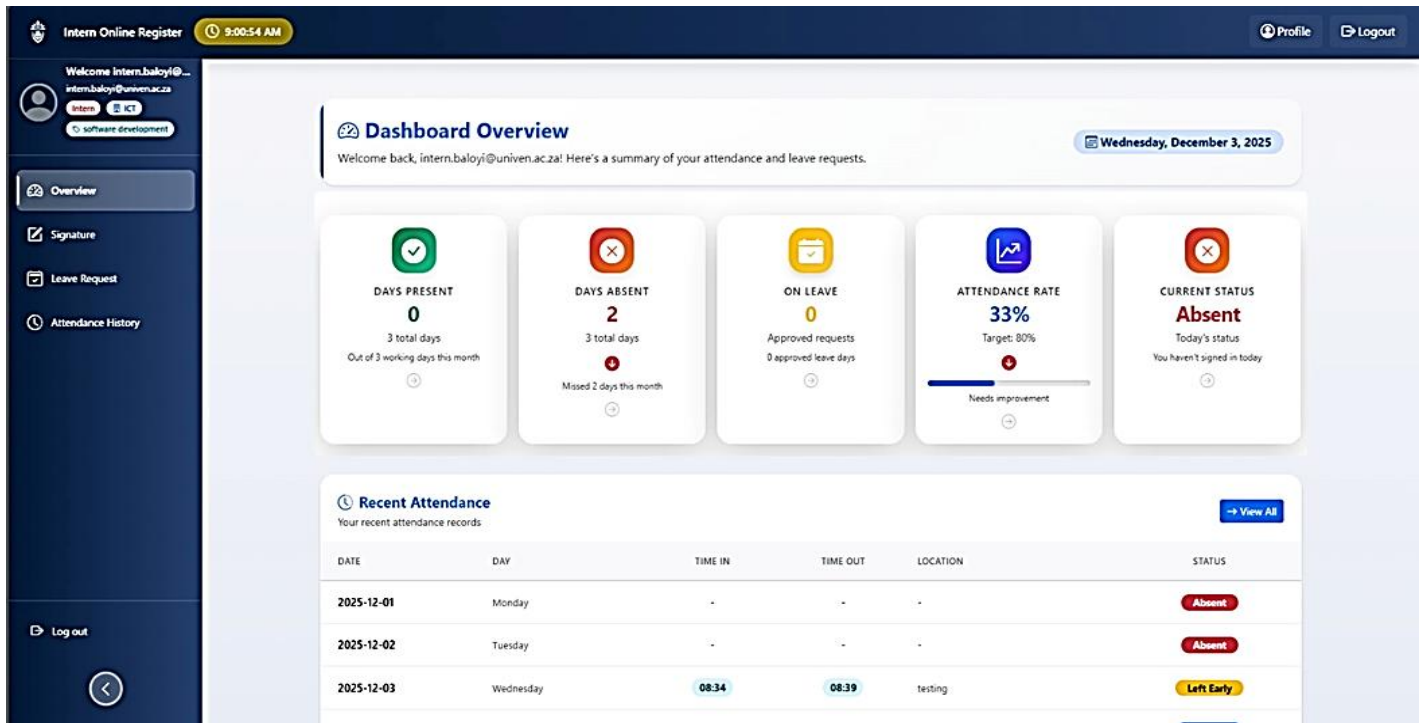
Supervisor Access (Leave)

- Approved leave
- Decline leave (must provide comment)
- Mark leaves as seen
- View attachment

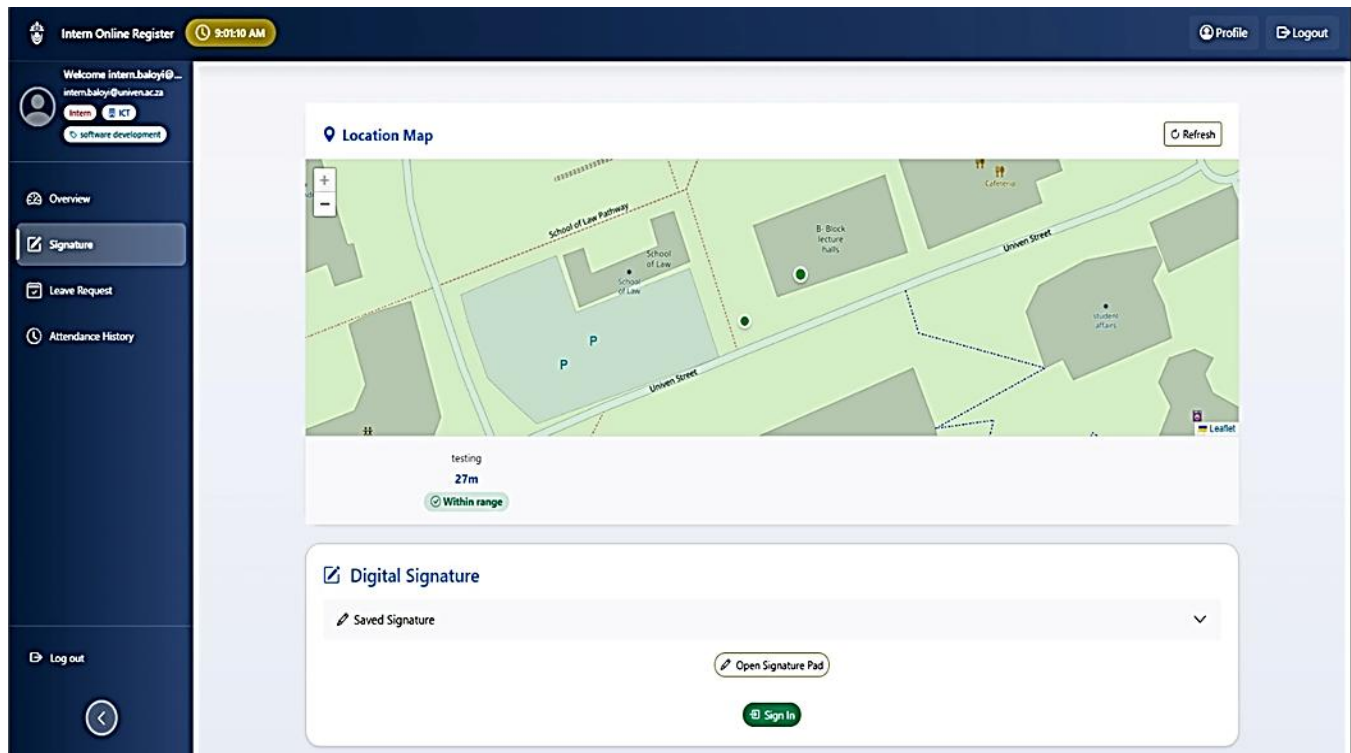
Admin Access (Leave)

- Approve/decline any departmental intern
- View full leave history
- Generate departmental leave reports

(G) Intern Interface (Dashboard)



1) Attendance (Sign In / Sign Out)



Database Tables Used:

- interns, attendance, leave_requests, users

Field Name	Data Type	Description / Source
Date	Date/Time	attendance.date (auto)
Time In / Time Out	Date/Time	attendance.time_in, attendance.time_out (auto)
GPS Location	Text / Double	attendance.location; latitude, longitude if captured
Status	Text	attendance.status (e.g., PRESENT/LATE/SIGNED_OUT)
Digital Signature	Drawing / Text	attendance.signature (longtext image payload) ≡
Intern Link	Hidden	attendance.intern_id → interns.intern_id

2) Leave Request

The screenshot shows a web application interface for an intern. The top navigation bar includes the user's name, email, and a clock showing 9:02:49 AM. The left sidebar contains a menu with options: Overview, Signature, Leave Request (highlighted), and Attendance History. The main content area is divided into two sections. The first section, 'Request Leave', contains a form with fields for 'Leave Type' (a dropdown menu), 'Start Date' (a date picker), and 'End Date' (a date picker). Below these is an 'Attach Document (Optional)' section with a 'Choose File' button and a 'No file chosen' message. A 'Submit Request' button is at the bottom right of the form. The second section, 'Leave Request History', shows a table with 2 requests found. The table has columns for ID, Type, Start, End, Attachment, Status, and Reason. The first request is 'SICK' status 'Approved'. The second request is 'ANNUAL' status 'Declined' with the reason 'Declined: no attachment'.

Request Leave

Fill in the form below to submit a leave request

Leave Type: Start Date: End Date:

Attach Document (Optional)

No file chosen

Optional: PDF, JPG, PNG, DOC, DOCX

Leave Request History

2 request(s) found

All Status

ID	Type	Start	End	Attachment	Status	Reason
1	SICK	Dec 4, 2025	Dec 11, 2025	4b263eaf-53e1-47e6-9470-2361d10ef4aa.pdf	Approved	-
2	ANNUAL	Dec 4, 2025	Dec 11, 2025		Declined	Declined: no attachment

Fields to be used:

Field Name	Data Type	Description / Source
Leave Type	Text	leave_requests.leave_type (Sick/Study/Annual/Family/etc.)
Start / End Date	Date	leave_requests.from_date, leave_requests.to_date
Reason	Text	leave_requests.reason
Attachment	File	leave_requests.attachment_path (stored reference)
Status	Text	leave_requests.status (Pending/Approved/Declined)
Intern Link	Hidden	leave_requests.intern_id → interns.intern_id

3) My Profile


Intern Online Register
8:47:33 AM
Profile
Logout

Settings

 Profile

 Change Password

[← Back to Dashboard](#)

Profile Information

First Name *

Surname

Email Address *

Role

Cancel

Update Profile

Data base table used:

Field Name	Data Type	Description / Source
Name / Email	Text	interns.name, interns.email (display); may sync with users.name, users.email
Department	Lookup	interns.department_id → departments.department_id
Supervisor	Lookup	interns.supervisor_id → supervisors.supervisor_id
Field	Text	interns.field
Employer	text	Intern employer
ID Number	Text	interns.id_number
Start / End Date	Date	interns.start_date, interns.end_date
Phone / Avatar (optional)	Text	users.phone, users.avatar_url (if you surface user fields)

(H) Supervisor Interface

The screenshot displays the Supervisor Interface dashboard. At the top, a dark blue header contains the text "Intern Online Register" and a clock showing "8:57:25 AM". On the right of the header are links for "Profile" and "Logout". A left sidebar menu includes "Dashboard", "Interns", "Leave Status", "History", and "Reports". The main content area starts with a "Dashboard Overview" section, which includes a welcome message and a date indicator for "Wednesday, December 3, 2025". Below this are four cards representing key metrics: "TOTAL INTERNS" (0), "PRESENT TODAY" (0), "ON LEAVE" (0), and "ABSENT" (0). Each card has a "Click to view details" link. The next section is "Intern Attendance Summary", which includes a table with columns for "INTERN", "DEPARTMENT", "PRESENT", "ABSENT", "ON LEAVE", and "% ATTENDANCE %". The table shows one intern, "intern baloyi", with 0 present, 0 absent, 0 on leave, and 0% attendance. The final section is "Recent Leave Requests", which includes a table with columns for "INTERN", "DEPARTMENT", "FIELD", "START DATE", "END DATE", "REASON", and "STATUS".

Dashboard Overview

Welcome back, muthuhadiniramavhunga@univen.ac.za! Here's a quick summary of your interns.

Wednesday, December 3, 2025

TOTAL INTERNS
0
Click to view details

PRESENT TODAY
0
Click to view details

ON LEAVE
0
Click to view details

ABSENT
0
Click to view details

Intern Attendance Summary
Track attendance records for all interns

Total: 1 interns

INTERN	DEPARTMENT	PRESENT	ABSENT	ON LEAVE	% ATTENDANCE %
intern baloyi intern.baloyi@univen.ac.za	ICT	0	0	0	0%

Recent Leave Requests
Recent leave requests from your interns

View All

INTERN	DEPARTMENT	FIELD	START DATE	END DATE	REASON	STATUS
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Database Tables Used:

- supervisors, interns, attendance, leave_requests, users (for contact details)

5) Supervisor Profile

 Intern Online Register 8:47:33 AM

[Profile](#) [Logout](#)

Settings

[Profile](#)

[Change Password](#)

[← Back to Dashboard](#)

Profile Information

First Name *

Surname

Email Address *

Role

SUPER_ADMIN

Cancel

Update Profile

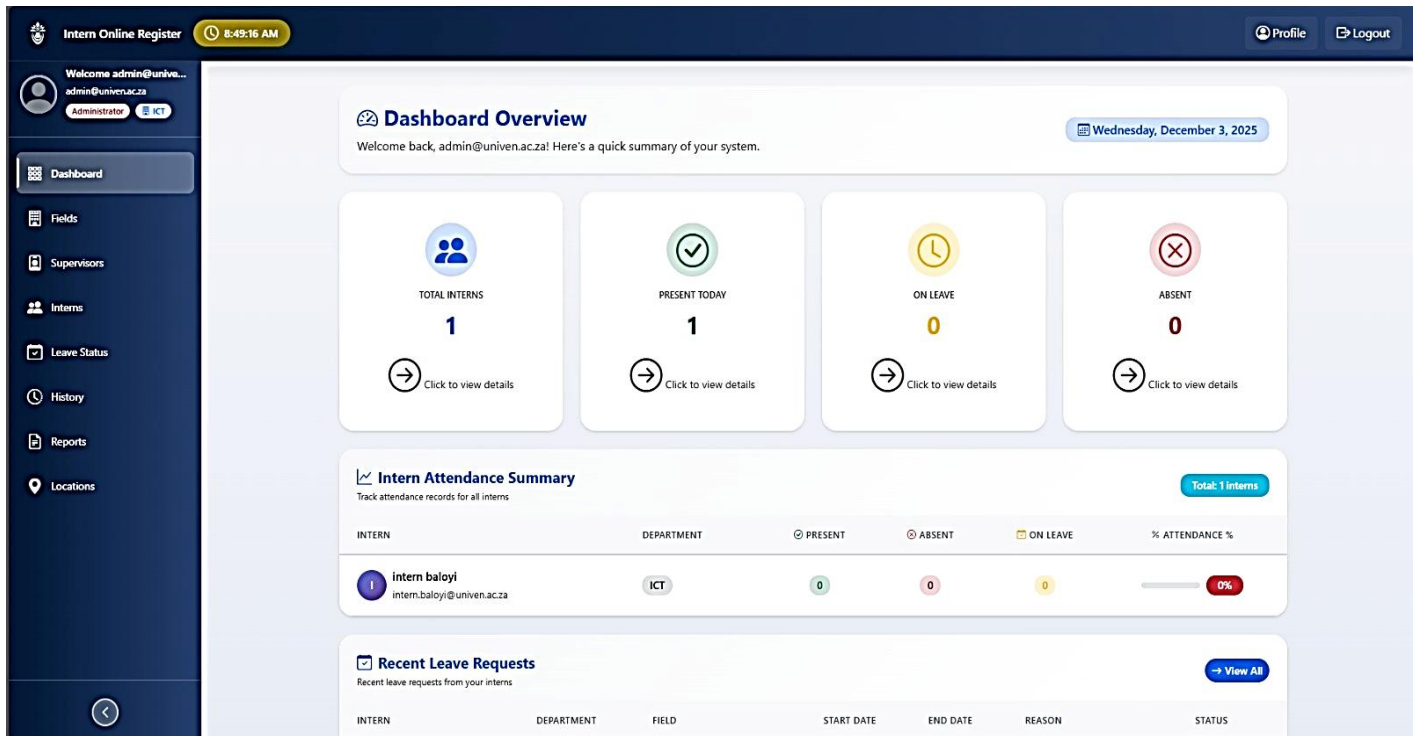
NB:

- Supervisor can update personal details; password change follows user account policy.

Fields to be used:

Field Name	Data Type	Description
Full Names	Text	supervisors.name (display & edit)
Email	Text	supervisors.email
Department	Lookup	supervisors.department_id → departments.department_id
Field (Specialization)	Text	supervisors.field
Account Metadata	Read-only	From users (e.g., last_login_at, phone, signature if used)

(I) Administrator Interface (Department Admin)



Database Tables Used:

- supervisors, fields, departments, users (for resets), admins (profile info)

1) Manage Supervisors

Manage Supervisors
Create, view, and manage all supervisors.

[Create New Supervisor](#)

Recently Added Supervisors - Not Logged In

Note: The following supervisors have been added to the system but have not logged in yet. Consider sending them their login credentials.

Search by name or email... All Departments All Fields Reset Filters

Showing 1 - 1 of 1 supervisor(s) Total: 1 Items per page: 25

Name	Email	Department	Field	Added Date	Actions
muthuhadini Ramavhunga	muthuhadiniramavhunga@univen.ac.za	ICT	software development	12/3/25, 8:29 AM	[Edit] [Delete]

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All Supervisors (1)
Main Supervisor Table for All Supervisors That Once Logged In to the System

Name/Email	Department	Field	Status
------------	------------	-------	--------

Fields to be used:

Field Name	Data Type	Description / Source
Name / Email	Text	supervisors.name, supervisors.email
Department	Lookup	supervisors.department_id → departments
Field (Specialization)	Text	supervisors.field (text column in schema)
Created / Updated	Date/Time	supervisors.created_at, supervisors.updated_at
Reset Password	Action	Target users' row of the supervisor account; issue reset token

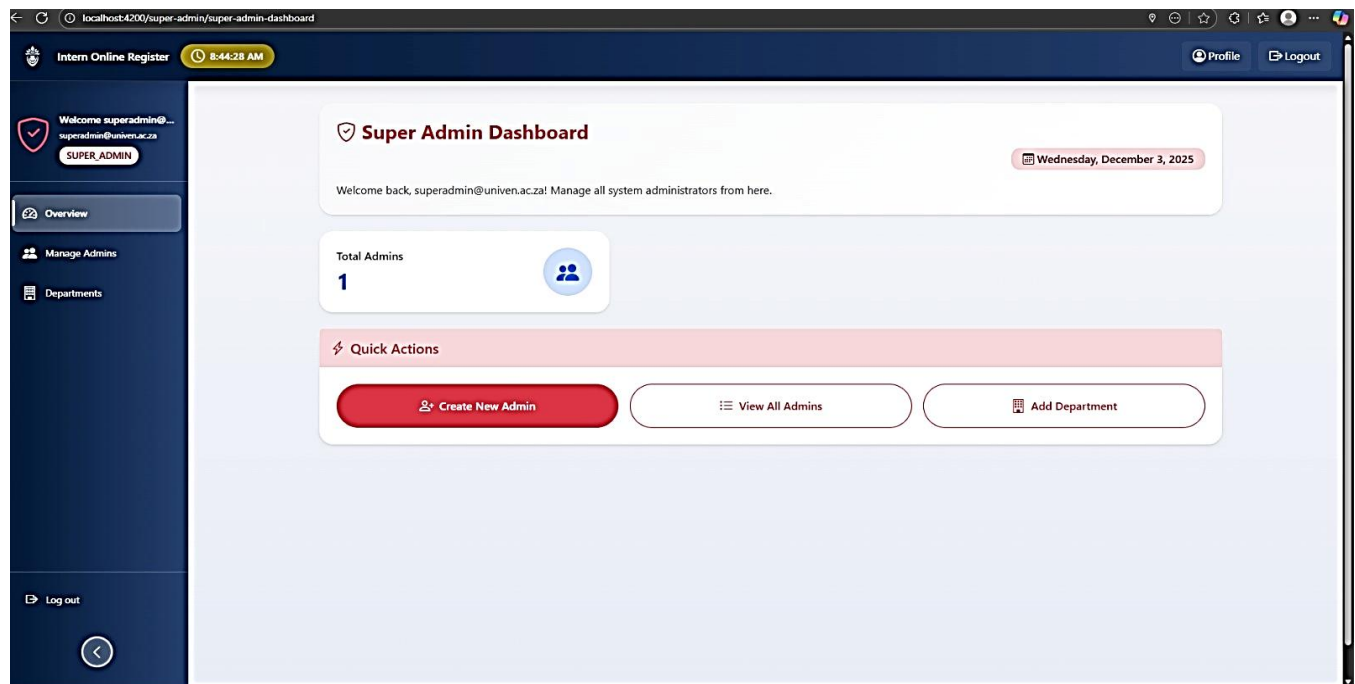
2) Manage Fields

The screenshot shows the 'Manage Department Fields' page. The top header includes 'Intern Online Register', the time '8:49:33 AM', and links for 'Profile' and 'Logout'. The left sidebar contains navigation links: 'Dashboard', 'Fields' (active), 'Supervisors', 'Interns', 'Leave Status', 'History', 'Reports', and 'Locations'. The main content area has a title 'Manage Department Fields' with a subtitle 'Add, edit, and deactivate fields for any department'. Below this is a section 'Add Field to Department' with a dropdown for 'Department' (set to 'ICT') and a text input for 'Field Name'. A message states: 'You can only manage fields for your assigned department: ICT'. An 'Add Field' button is present. Below this is a table titled 'ICT Department Fields' showing one field: 'software development' with edit and delete icons. The table has columns 'Field Name' and 'Actions'.

Fields to be used:

Field Name	Data Type	Description / Source
Field Name	Text	fields.name
Department	Lookup	fields.department_id → departments.department_id
Active	Boolean	fields.active
Created / Updated	Date/Time	fields.created_at, fields.updated_at

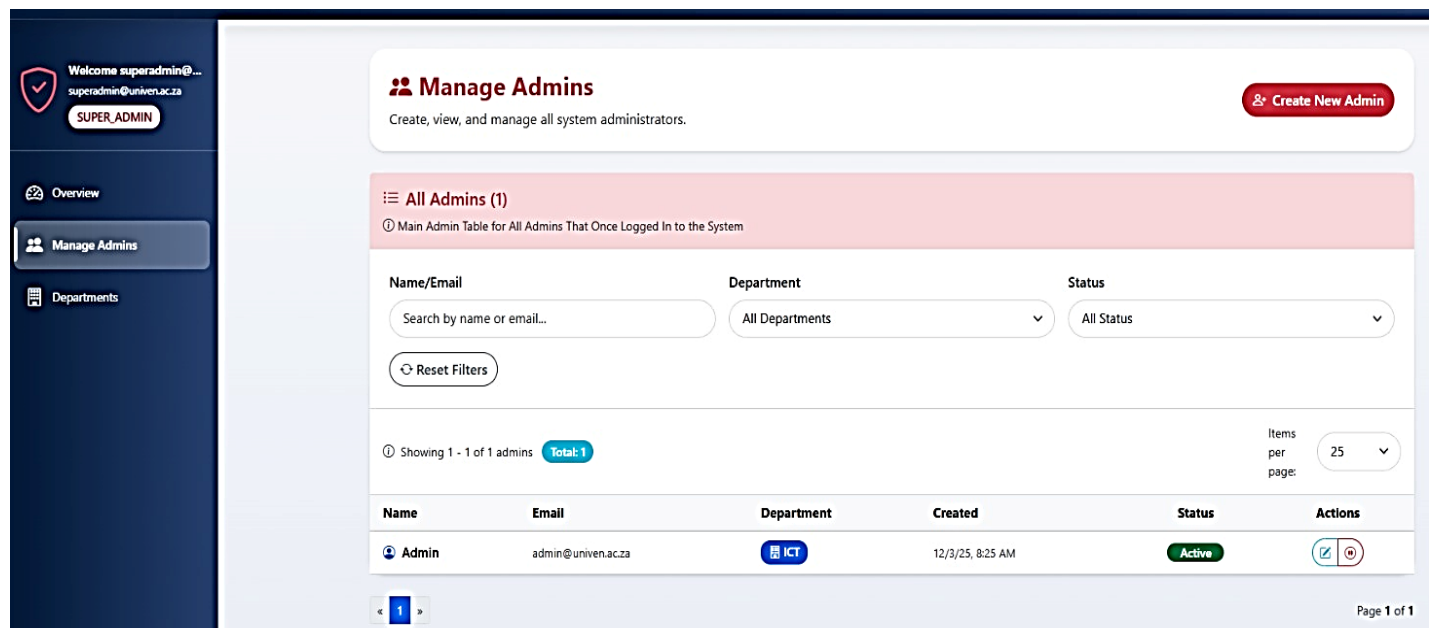
(J) Super Administrator Interface (University-Wide)



Database Tables Used:

- admins, departments, super_admins, users

1). Manage Admins



Fields to be used:

Field Name	Data Type	Description / Source
Name / Email	Text	admins.name, admins.email
Department	Lookup	admins.department_id → departments.department_id
Active	Boolean	admins.active
Created / Updated	Date/Time	admins.created_at, admins.updated_at

2) Manage Departments

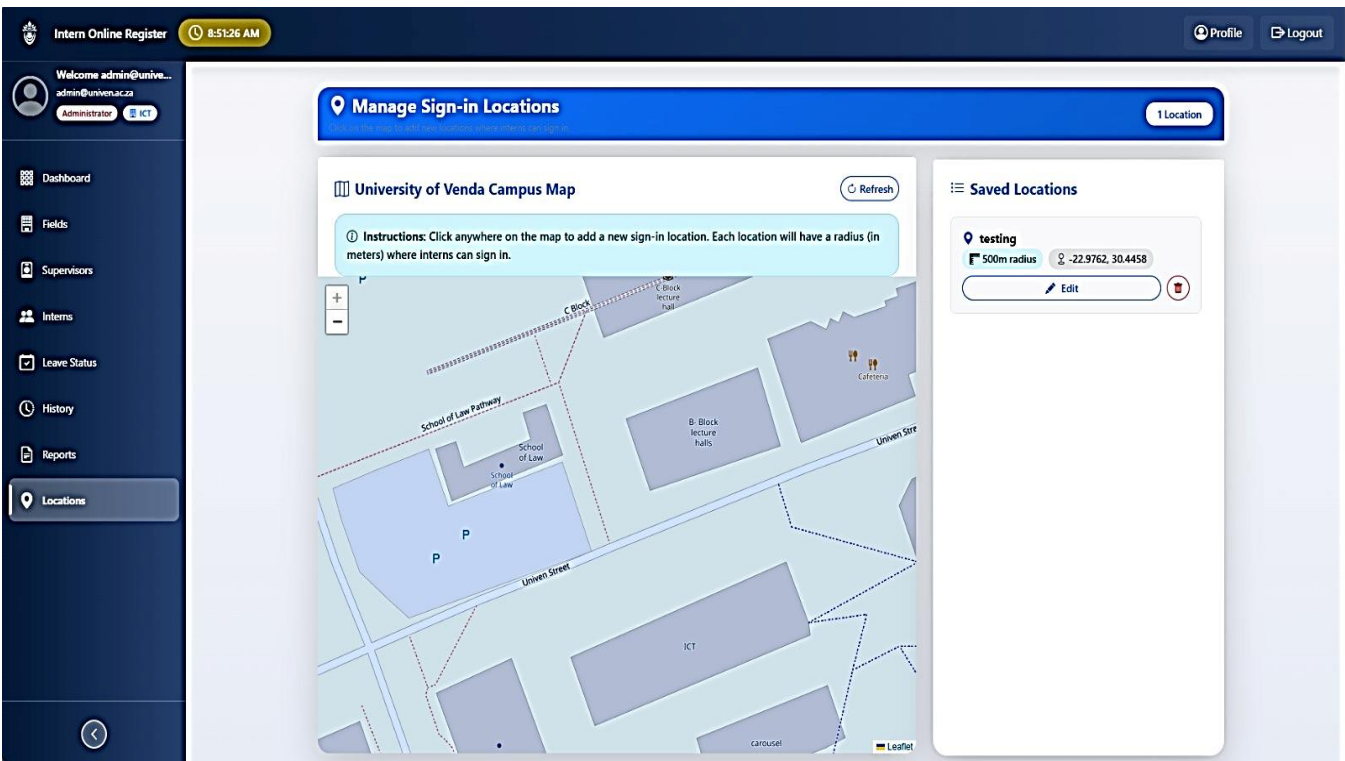
The screenshot shows the 'Manage Departments' page. At the top, a dark blue header contains the text 'Intern Online Register', a clock showing '8:45:24 AM', and links for 'Profile' and 'Logout'. A left sidebar contains a 'Welcome' message for 'superadmin@univen.ac.za' with the role 'SUPER_ADMIN', and navigation links for 'Overview', 'Manage Admins', and 'Departments' (which is highlighted). The main content area has a title 'Manage Departments' with a subtitle 'Create, view, edit, and delete departments.' and a red 'Add New Department' button. Below this is a table titled 'All Departments (1)'. The table has five columns: 'ID', 'Department Name', 'Fields Count', 'Status', and 'Actions'. It contains one row for the 'ICT' department, which has 1 field and is 'Active'. The 'Actions' column for this row contains edit and delete icons.

ID	Department Name	Fields Count	Status	Actions
#1	ICT	1 field	Active	

Fields to be used:

Field Name	Data Type	Description / Source
Department Name	Text	departments.name
Active	Boolean	departments.active
Created / Updated	Date/Time	departments.created_at, departments.updated_at

(K) Location Management (Geofencing)



Database Tables Used:

- locations

Fields to be used:

Field Name	Data Type	Description / Source
Name	Text	locations.name (site/campus)
Latitude / Longitude	Double	locations.latitude, locations.longitude
Radius (meters)	Integer	locations.radius (geo-fence)
Active	Boolean	locations.active
Created / Updated	Date/Time	locations.created_at, locations.updated_at

(L) Notification Preferences

Database Tables Used: notification_preferences, users

Fields to be used:

Field Name	Data Type	Description / Source
Attendance Alerts	Boolean	notification_preferences.email_attendance_alerts
Leave Updates	Boolean	notification_preferences.email_leave_updates
Frequency	Enum	notification_preferences.frequency (DAILY/INSTANT/WEEKLY)
User Link	Hidden	notification_preferences.user_id → users.id

5.1.3. Integration interface:

- a) ITS integration: Extract HR/organizational data (Departments, Staff/Assignments) for provisioning and alignment; ITS-sourced master data remains read-only in IRMS.
- b) EDRMS integration: Store supporting documents (leave attachments, reports) and key attendance extracts for retention and audit.
- c) Notifications (Email) [**Optional**] Leave decisions and periodic summaries to relevant roles.

6. Non-Functional Requirements

6.1 Performance Requirements

- **Response Time:**

The system dashboard and main navigation pages shall load within 2 seconds over a standard broadband connection (4G/Fiber).

- **Transaction Processing:**

Sign-in/Sign-out requests shall be processed and confirmed within 1 second to avoid queuing during peak hours (08:00 AM – 09:00 AM).

- Queries for attendance or leave reports spanning up to one year of data shall execute and return results within 5 seconds.
- The system shall provide real-time updates for attendance records, dashboards, and reporting data without requiring the user to reload the page. Updates should be pushed automatically within **1 second** of the change occurring.

6.2 Reliability & Availability

- **Availability:**

The system shall maintain an uptime of 99.9% during core business hours (07:00 AM – 18:00 PM, Monday–Friday).

- **Data Integrity:**

Data must auto-save every time a change is made to prevent loss of entries.

- **Fault Tolerance:**

The system shall gracefully handle temporary client-side network failures (e.g., allow offline queuing of Sign-in events until reconnection).

6.3 Scalability

- The system must scale to accommodate future expansion, including additional faculties, interns, or staff.
- Database structure must support growth to **over 1 million attendance records** without performance degradation.
- New modules (e.g., Leave Management, Biometrics) should be integrable without redesigning the entire system.

6.4. Usability

- New interns shall be able to complete core tasks (Sign In, Sign Out, Request Leave) with under 10 minutes of training or via a one-page guide.
- The Intern Dashboard must be fully functional and responsive on mobile devices with screen widths as small as 320px.
- system interface must be simple and intuitive for Interns, Supervisors, and Admins.
- mandatory fields must be marked clearly using an asterisk
- The system must be accessible via desktop, laptop, and tablet.
- Font sizes, colors, and layout must comply with the university's UI/UX standard guidelines.
- Users must be able to download reports in common formats (PDF and Excel) without needing third-party tools.

6.5 Security & Privacy

- All users must authenticate using your **univen Staff email**, in alignment with university login policies.
- User roles (Intern, Supervisor, Admin, Super admin) must strictly control visibility and access permissions.
- Passwords and sensitive information must never be stored in plain text.
- Audit logs must record all modifications to attendance entries and report generation activities.
- The system must comply with **University Data Protection and Privacy Policies**.
- User sessions must automatically expire after 30 minutes of inactivity.
- The system shall collect location data only at the moment of Sign-in/Sign-out and shall not continuously track user movement.
- The system shall prevent a user from maintaining more than one active session simultaneously. If the same user attempts to log in from another device or browser, the previous session shall automatically terminate.

6.6 Maintainability

- System configuration and codebase must be modular to allow future expansion (e.g., adding biometric integration).
- All interfaces must use identifiable and consistent naming conventions for easier maintenance.
- Error logs, system logs, and audit logs must be easily retrievable by ICT support.
- System must support REST API integration to allow interoperability with other university systems.

6.7. Compatibility Requirements

- The system must run on all modern browsers: Chrome, Edge, Firefox, Safari.
- It must work with both Windows and macOS environments.
- Reports must open on standard software such as Adobe Reader and Microsoft Excel.

6.8. Backup & Disaster Recovery Requirements

- Attendance data must be backed up **daily**.
- Backups should be stored in a secure off-site location in compliance with ICT policy.
- In case of major data loss, system recovery should restore data to the last backup point with minimal downtime.

6.9 Audit & Compliance Requirements

- **Every attendance modification, approval, or rejection must be logged with:**
 - User ID
 - Timestamp
 - Action performed
 - Old value and new value
- **The system must comply with:**
 - ICT Governance & Security Standards
 - University Data Management Policies
 - Audit and Compliance requirements for internal reviews

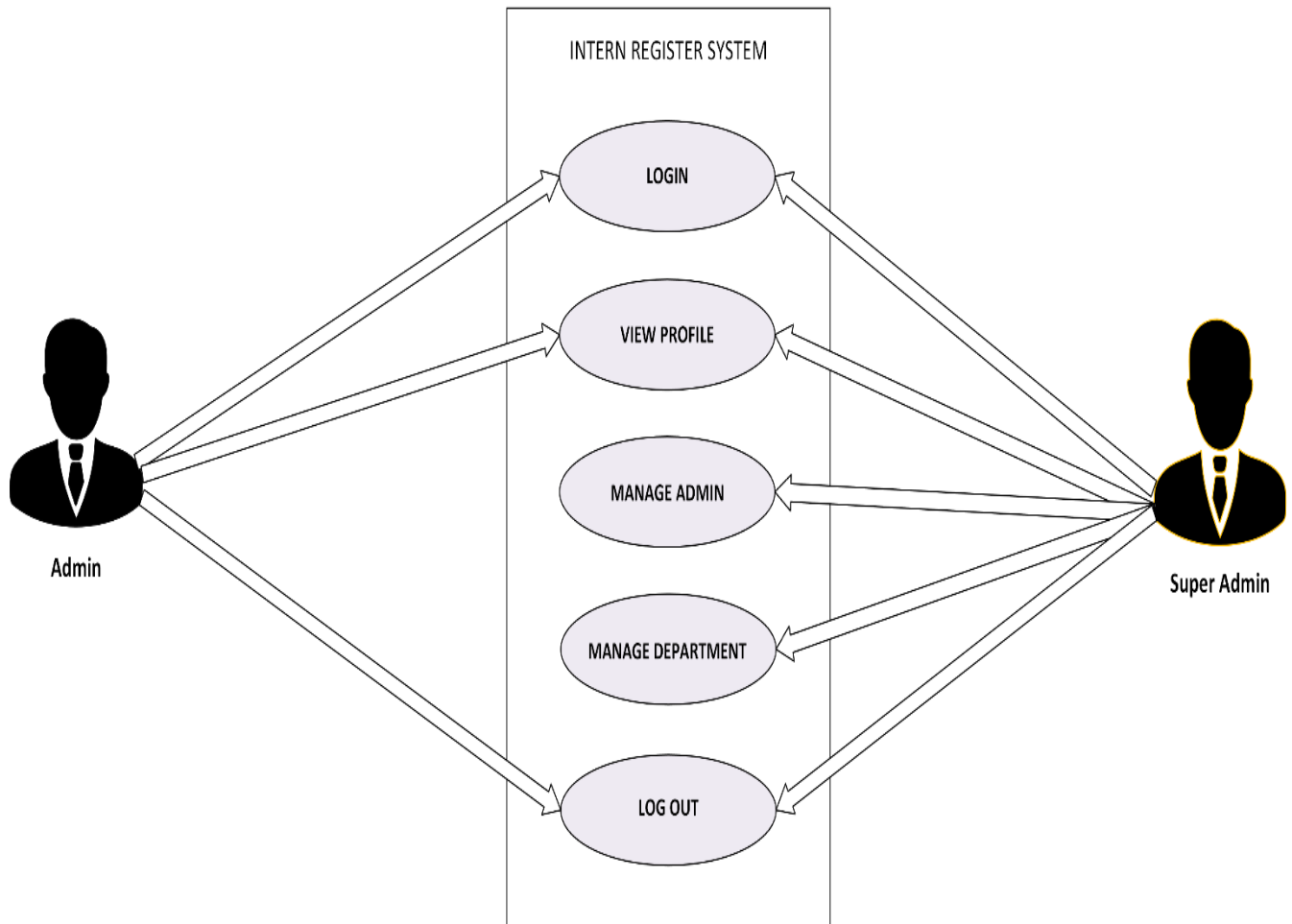
7. Use cases and workflow Diagrams

7.1. Use case model

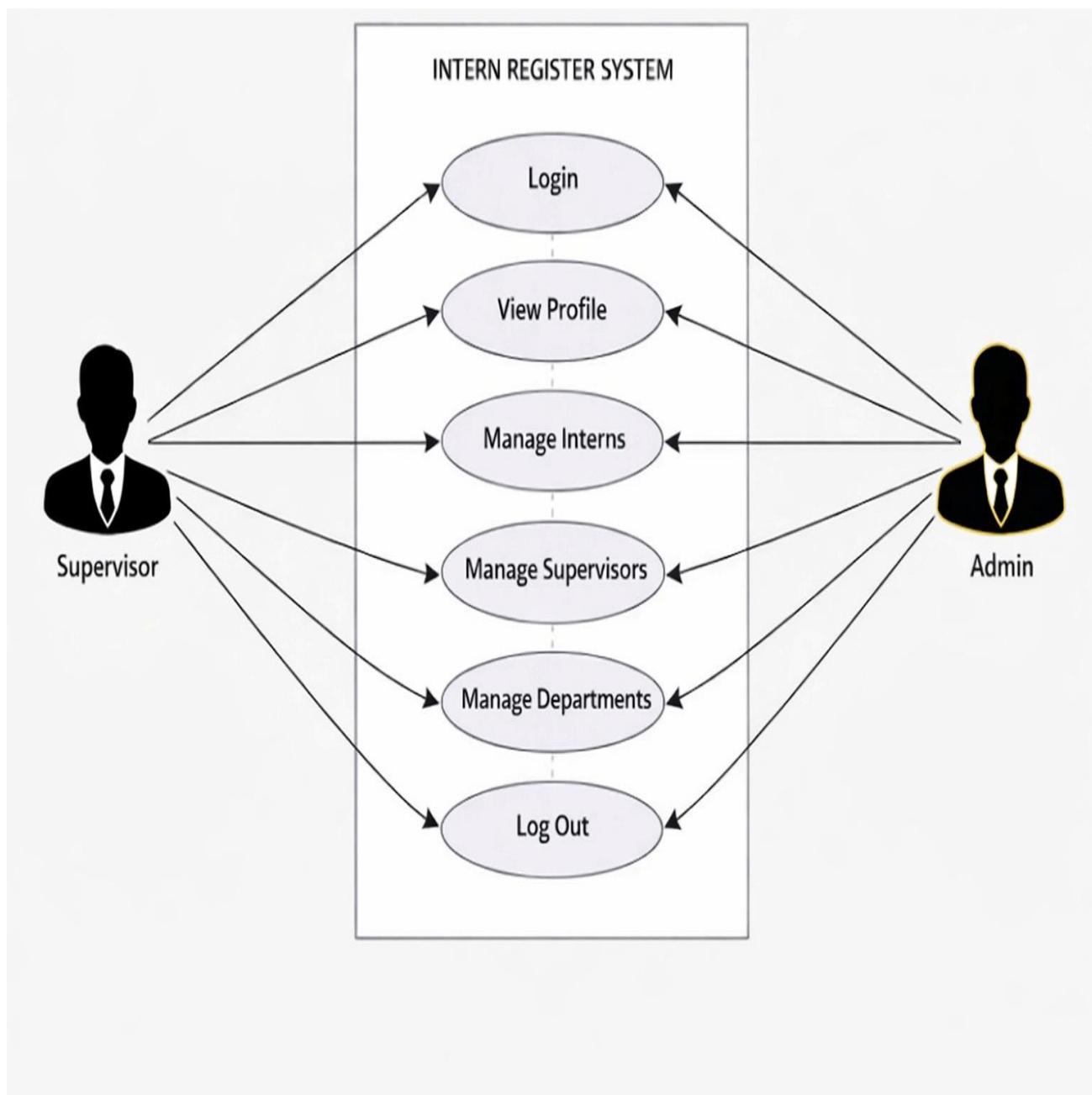
7.1.1. Actors

Actor	Description	Key System Actions
Super Admin	Highest-level system user with full control over configuration and user management.	Manage Admins, Manage Departments, View System Reports.
Administrator	Manages a specific department and its users.	Manage Supervisors & Interns, Approve Leave, View Department Reports.
Supervisor	Oversees assigned interns and monitors their attendance.	View Attendance, Approve Leave, Update Intern Details.
Intern	Records daily attendance and requests leave.	Sign In/Out, Submit Leave, View Personal Attendance.

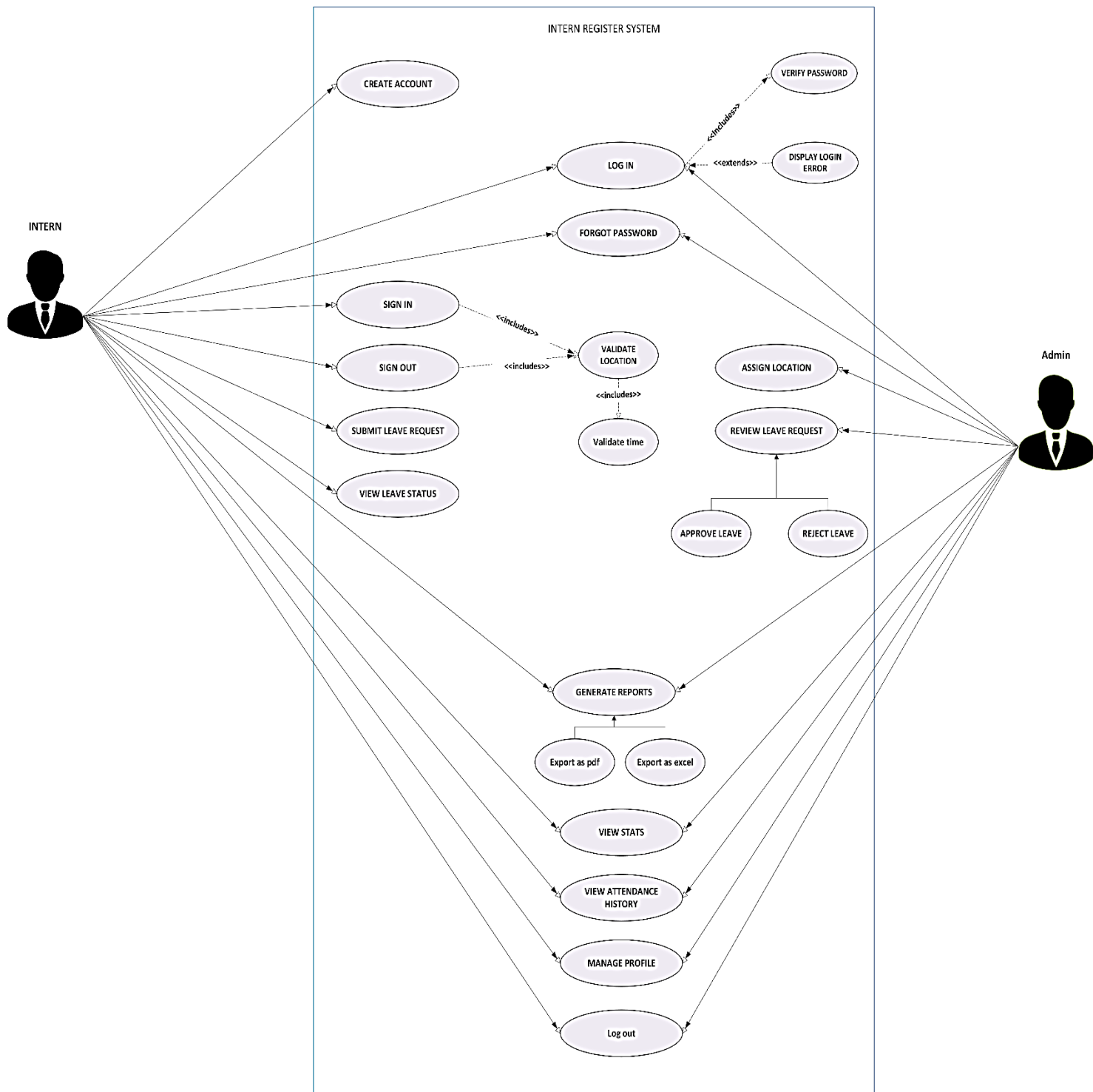
7.1.1. super admin and admin use case diagram



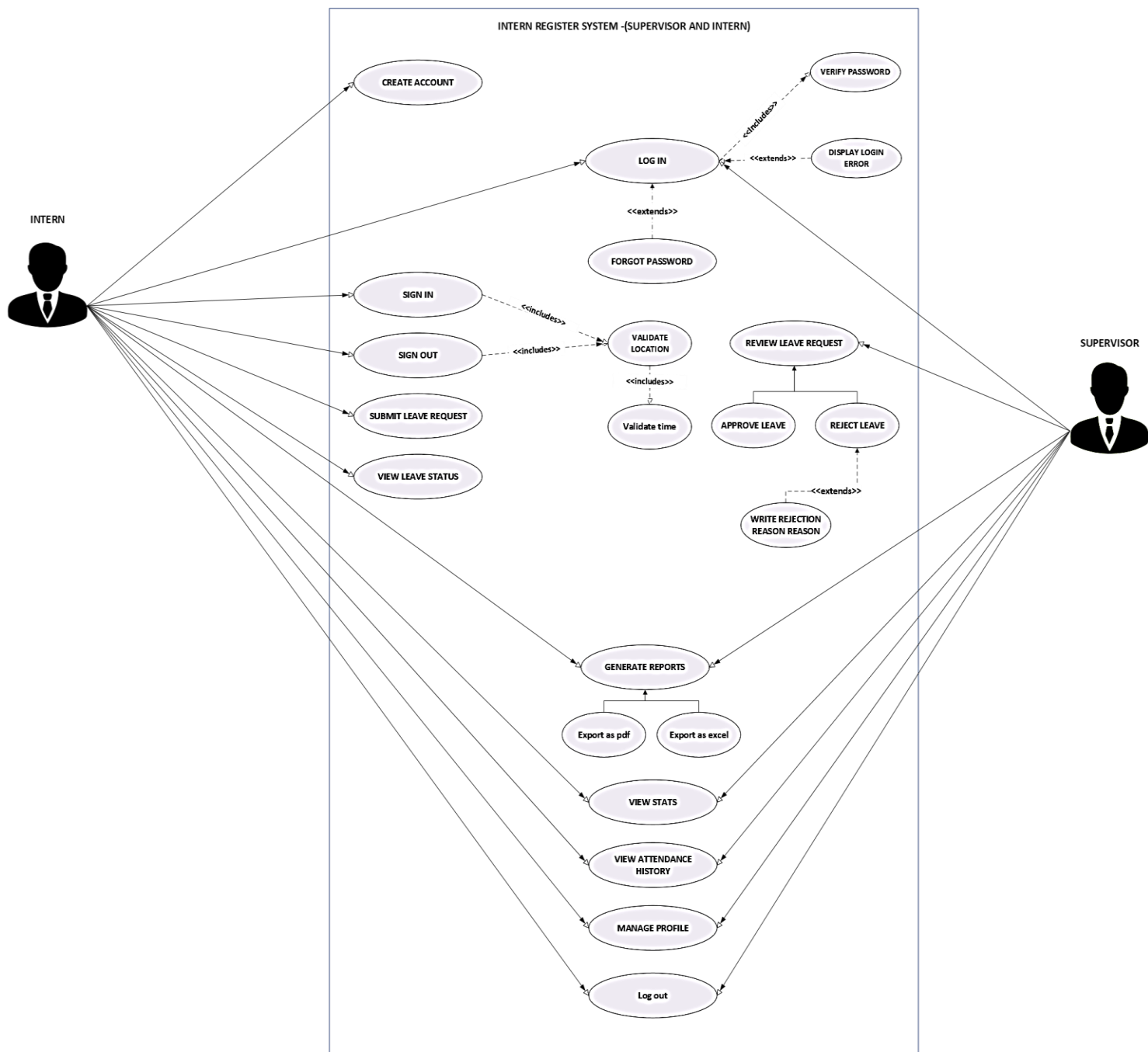
7.1.2. admin and supervisor dashboard



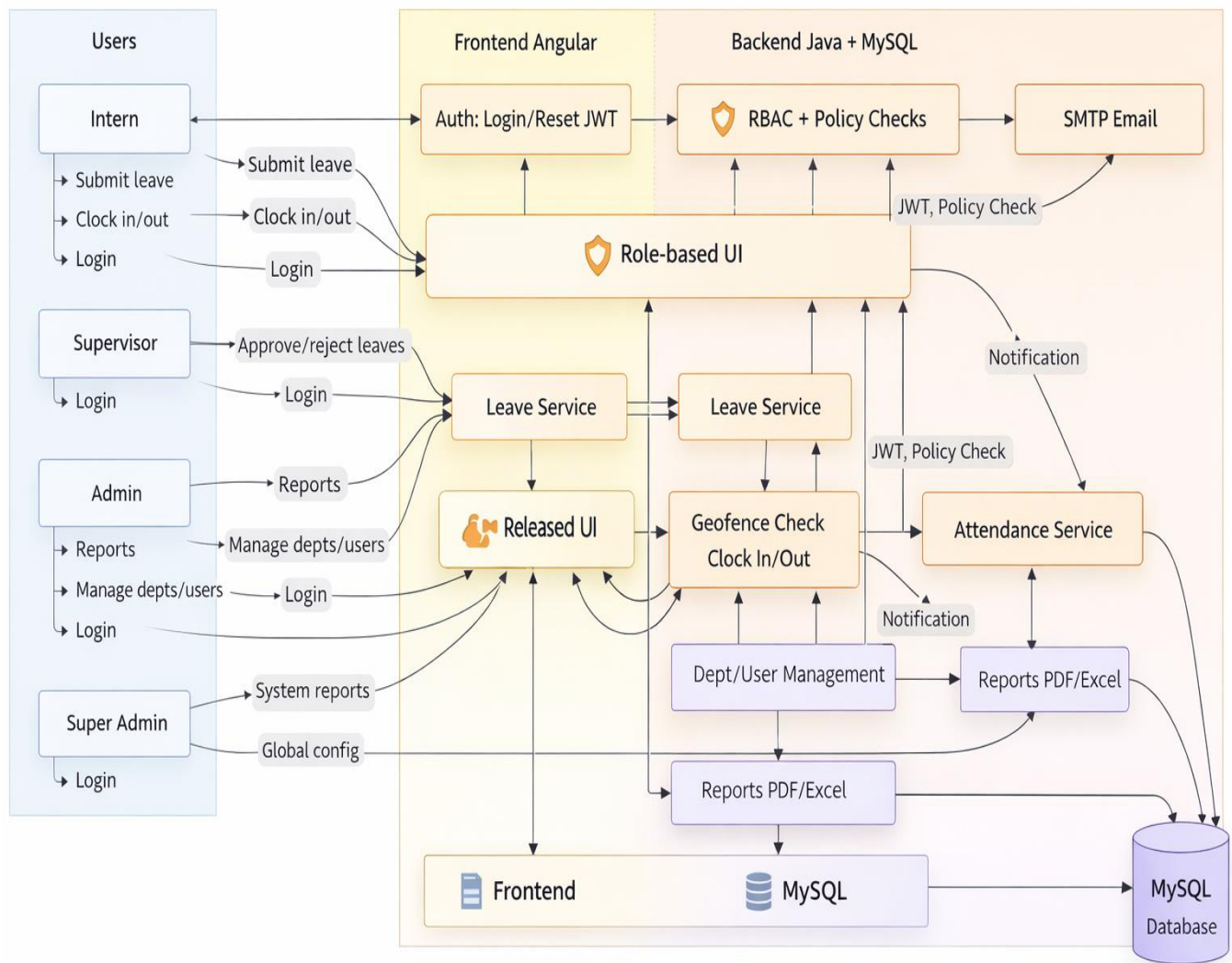
7.1.3. admin and intern dashboard



7.1.4. Supervisor and Intern dashboard



7.2. workflow diagram



11. Recommendations

Based on the analysis of current challenges and the functional requirements outlined in this document, the following recommendations are proposed to ensure the successful implementation and long-term sustainability of the Intern Register Management System (IRMS):

1) **Adopt a phased rollout strategy**

Implement the system across departments in structured phases to minimize disruption, allow early feedback, and progressively refine system functionality before institution-wide deployment.

2) **Conduct training and capacity-building sessions**

Provide targeted training for Interns, Supervisors, Admins, and HR personnel to ensure effective system adoption. Emphasis should be placed on geofencing rules, leave workflows, and the reporting features.

3) **Enforce data governance and standardization**

Establish clear data-entry standards (e.g., naming conventions, field mappings, department codes) to enhance data accuracy, audit trails, and report consistency.

4) **Strengthen device and browser compatibility testing**

Ensure the system is fully responsive and tested across both desktop and mobile browsers to support interns who rely primarily on mobile devices for attendance logging.

5) **Implement continuous monitoring and system performance tracking**

Set up logging, alerting, and routine audits to monitor system uptime, geolocation accuracy, and attendance anomalies, ensuring reliability and early issue detection.

6) **Integrate IRMS with future institutional systems**

Design the system to be *integration-ready* for future extensions such as HR systems, payroll processing tools, or institutional authentication services (e.g., SSO).

7) **Formalize a support and maintenance plan**

Assign internal technical support, define escalation procedures, and maintain documentation to ensure seamless operation after deployment.

12. Conclusion

The Intern Register Management System (IRMS) provides a comprehensive, modernized solution to address longstanding administrative challenges in intern attendance tracking, leave management, and departmental reporting. By replacing manual logbooks and fragmented processes with a secure, geolocation-enabled digital platform, IRMS significantly enhances accuracy, transparency, and efficiency across the University of Venda.

The system aligns directly with the institution's digital transformation objectives, improving service delivery and reducing the administrative workload for HR personnel, supervisors, and departmental administrators. With capabilities such as automated attendance verification, streamlined leave workflows, audit trails, and role-based dashboards, IRMS positions the University to better manage its internship programmes and support evidence-based decision-making.

Successful implementation of this system will not only modernize day-to-day operations but also reinforce accountability, improve stipend processing, and enhance the overall stakeholder experience. Adopting the recommendations outlined in this document will ensure that the IRMS is deployed effectively and sustainably, supporting long-term institutional goals and operational excellence.